

12	A balloon contains 2.0 moles of helium gas at a temperature of 27 °C. The molar mass of helium is 4.0 g mol⁻¹. What is the average translational kinetic energy of the helium atoms in the balloon?					
	A 5.6 × 10 ⁻²² J	B 6.2 × 10 ⁻²¹ J	C 30 J	D 7500 J		
	Answer: B					

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$$\text{Average K.E.} = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(27 + 273.15) = 6.2 \times 10^{-21} \text{ J}$$

13 A 350 cm³ cup of hot tea takes 30 minutes to cool down from an initial temperature of 90 °C